ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 2 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable sketches.

SECTION A

QUESTION 1

1.1 C ✓
1.2 C ✓
1.3 A ✓
1.4 D ✓
1.5 B ✓
1.6 D ✓
1.7 B ✓
1.8 C ✓
1.9 A ✓
1.10 B ✓

(10 × 1) [10]

QUESTION 2

2.1 False ✓
2.2 True ✓
2.3 True ✓
2.4 False ✓
2.5 True ✓
2.6 True ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 True ✓

(10 × 1) [10]

QUESTION 3

3.1 C ✓
3.2 H ✓
3.3 F ✓
3.4 D ✓
3.5 A ✓
3.6 J ✓
3.7 B ✓

3.8 G ✓

3.9 E ✓

3.10 I ✓

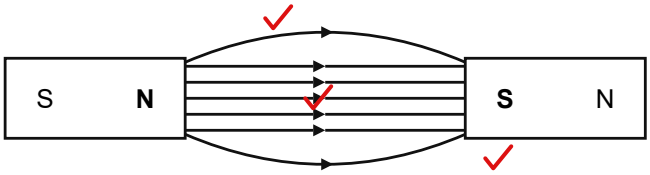
(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1



(Lines go straight across from N to S 1, curving outwards at the edges 1, arrows from N to S 1)

(3)

4.2 Any THREE ✓✓✓: increase the current; increase the number of turns; use a soft-iron core; wind the turns closer together (3)

4.3 The small Arduino current flows through the relay coil ✓, which becomes an electromagnet ✓ and pulls the armature / moves the contacts ✓, closing a separate circuit that carries the large motor current (the two circuits are isolated) ✓. (4)

4.4 The emf induced in a conductor is proportional to the rate at which it cuts (the rate of change of) magnetic flux ✓✓. Factors ✓✓✓: strength of the magnetic field; speed of movement (rate of change); number of turns on the coil; (length of conductor in the field) (5)

[15]

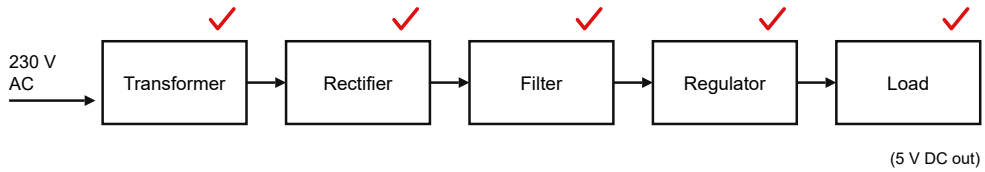
QUESTION 5

5.1.1 Step-down ✓ (1)

5.1.2 $V_s \div V_p = N_s \div N_p$ ✓ $\rightarrow V_s = 230 \times 60 \div 1\,150$ ✓ $= 12$ V ✓ (3)

5.1.3 $V_p I_p = V_s I_s$ ✓ $\rightarrow I_p = 12 \times 2 \div 230$ ✓ $= 0{,}10$ A ✓ (3)

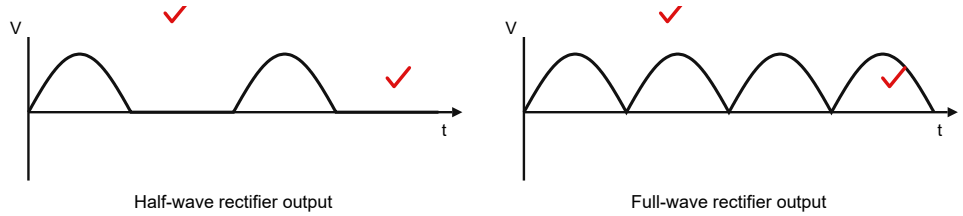
5.2



(Each of the five blocks in the correct order 1 and its function 1: transformer – steps 230 V AC down to a low AC voltage; rectifier – changes AC into pulsating DC; filter (smoothing capacitor) – smooths the pulsating DC and reduces ripple; regulator (e.g. 7805) – keeps the output at a constant 5 V; load – the robot circuit that uses the 5 V DC)

(10)

5.3



(Half-wave: only the positive half-cycles with gaps between them 2; full-wave: every half-cycle as a positive hump, no gaps 2)

(4)

[21]

QUESTION 6

6.1 $V_L = \sqrt{3} \times V_{ph}$ ✓ $= \sqrt{3} \times 230 = 398{,}37$ V (≈ 400 V) ✓ (2)

6.2 Any TWO ✓✓: more power with less copper; three-phase motors are self-starting, smaller and smoother; constant power delivery; two voltages available (230 V and 400 V) (2)

6.3.1 $C_T = 10 + 20 + 20$ ✓ $= 50\ \mu F$ ✓ (2)

6.3.2 $1/C_T = 1/10 + 1/20 + 1/20$ ✓ $= 0{,}2$ ✓; $C_T = 5\ \mu F$ ✓✓ (4)

6.3.3 $Q = C \times V$ ✓ $= 50 \times 10^{-6} \times 12 = 6 \times 10^{-4}$ C ($600\ \mu C$) ✓ (2)

[12]

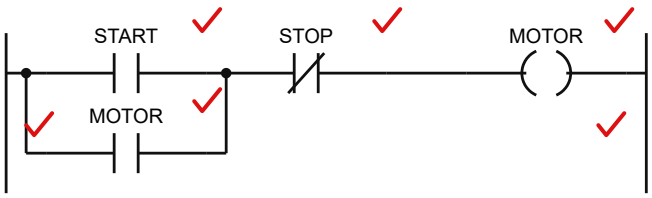
QUESTION 7

7.1 Any FOUR (part ✓ + function ✓): yoke / frame – supports the motor and carries the magnetic flux; field poles / field winding or permanent magnets – produce the magnetic field; armature – rotating coils on which the force acts; commutator – reverses the current in the armature coils every half turn so it keeps turning; brushes – carry current to the commutator; bearings / shaft – allow rotation and deliver the mechanical output (8)

7.2 A motor converts electrical energy into mechanical (rotational) energy ✓; a generator converts mechanical energy into electrical energy ✓. (2)

7.3 Power supply ✓; CPU / processor (with memory) ✓; input module ✓; output module ✓ (accept programming device as one) (4)

7.4



(START normally open 1, STOP normally closed in series 1, motor output coil 1, holding (seal-in) contact of the motor in parallel with START 2, both power rails and labels 1)

(6)

7.5 Any TWO ✓✓: easy to change the program instead of rewiring; less wiring and smaller panels; fault-finding with the programming software; timers and counters built in; reliable in harsh environments; can communicate with networks / SCADA (2)

[22]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE